



[최종 정리] LightGBM 산출물 (2차 전처리) ◦



Task :



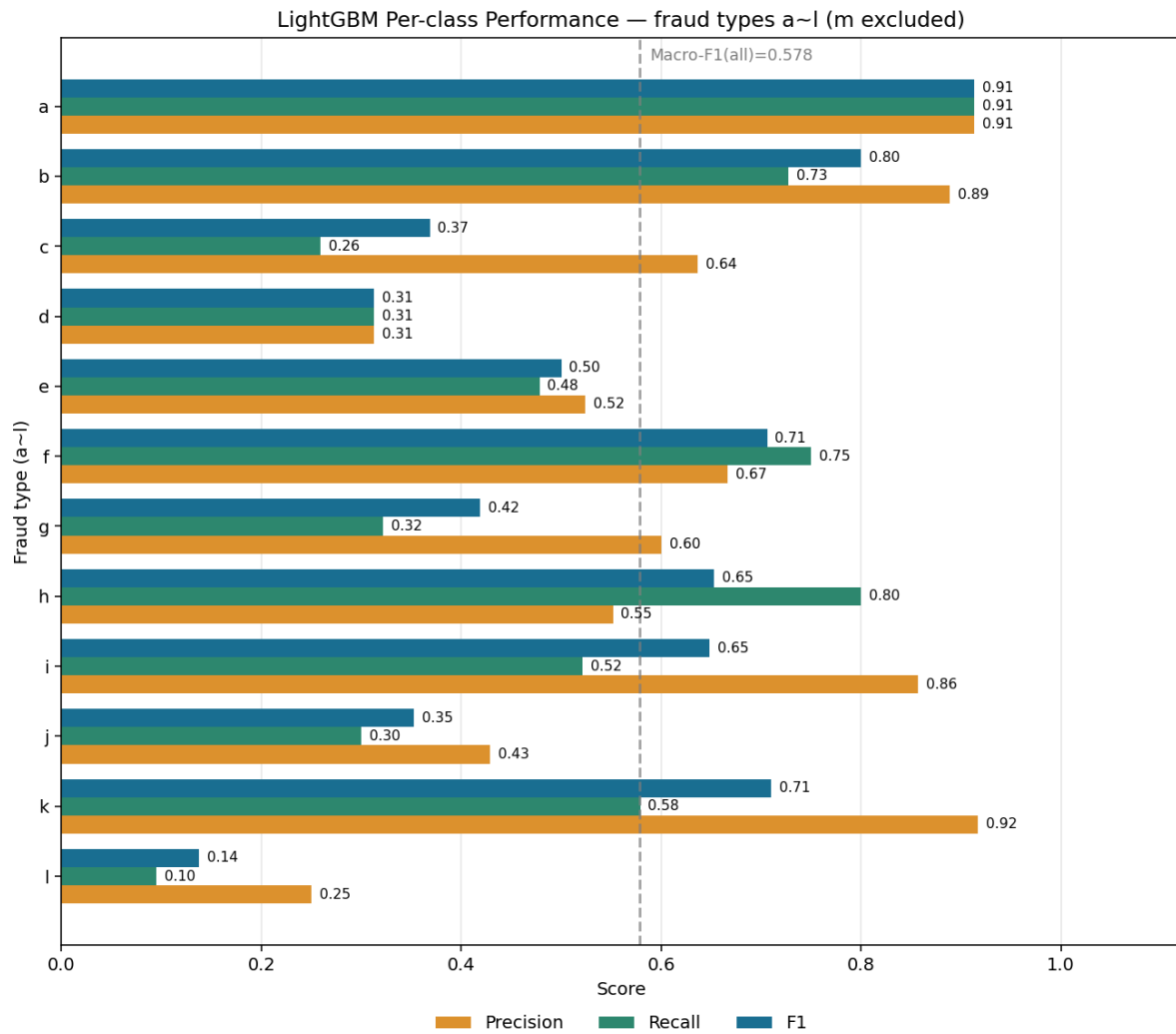
실행 및 진행 사항 정리

LightGBM 산출물 (2차 전처리)

2차 전처리에 정답 라벨 `is_fraud` 가 섞여 있어 제외하고 학습함

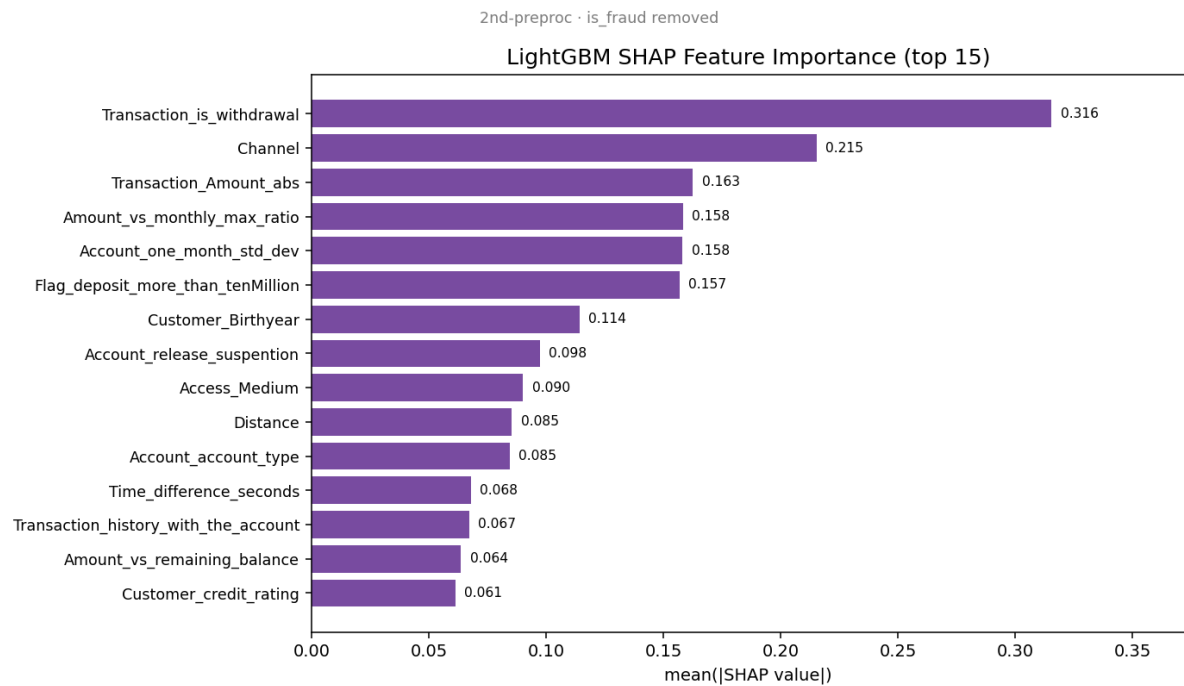
클래스별_성능

2nd-preproc · is_fraud removed



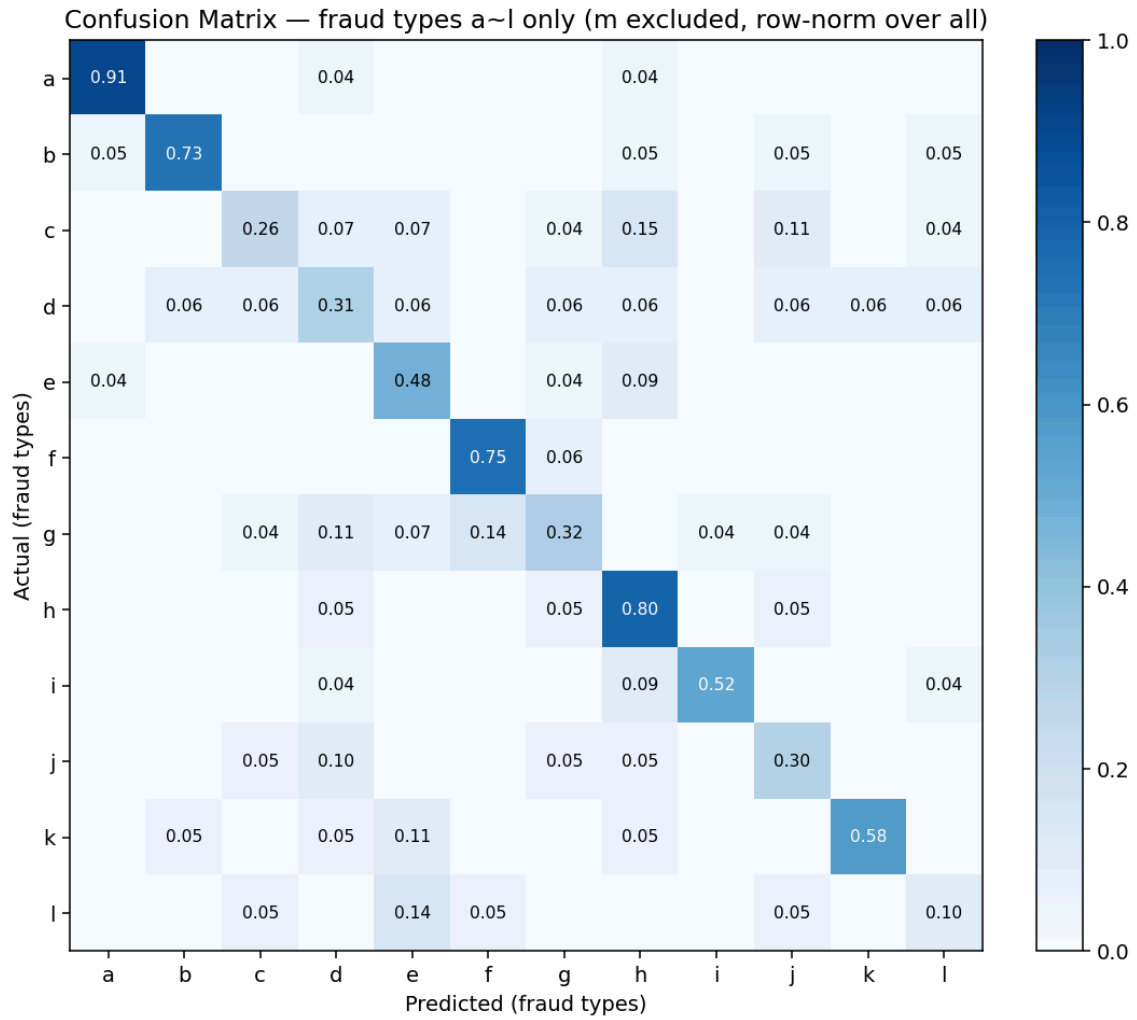
클래스	precision	recall	f1	support
a	0.913	0.913	0.913	23
b	0.8889	0.7273	0.8	22
c	0.6364	0.2593	0.3684	27
d	0.3125	0.3125	0.3125	16
e	0.5238	0.4783	0.5	23
f	0.6667	0.75	0.7059	16
g	0.6	0.3214	0.4186	28
h	0.5517	0.8	0.6531	20
i	0.8571	0.5217	0.6486	23
j	0.4286	0.3	0.3529	20
k	0.9167	0.5789	0.7097	19
l	0.25	0.0952	0.1379	21
m	0.9973	0.9998	0.9986	23602
macro avg	0.6571	0.5429	0.5784	23860
weighted avg	0.9934	0.9944	0.9936	23860

SHAP



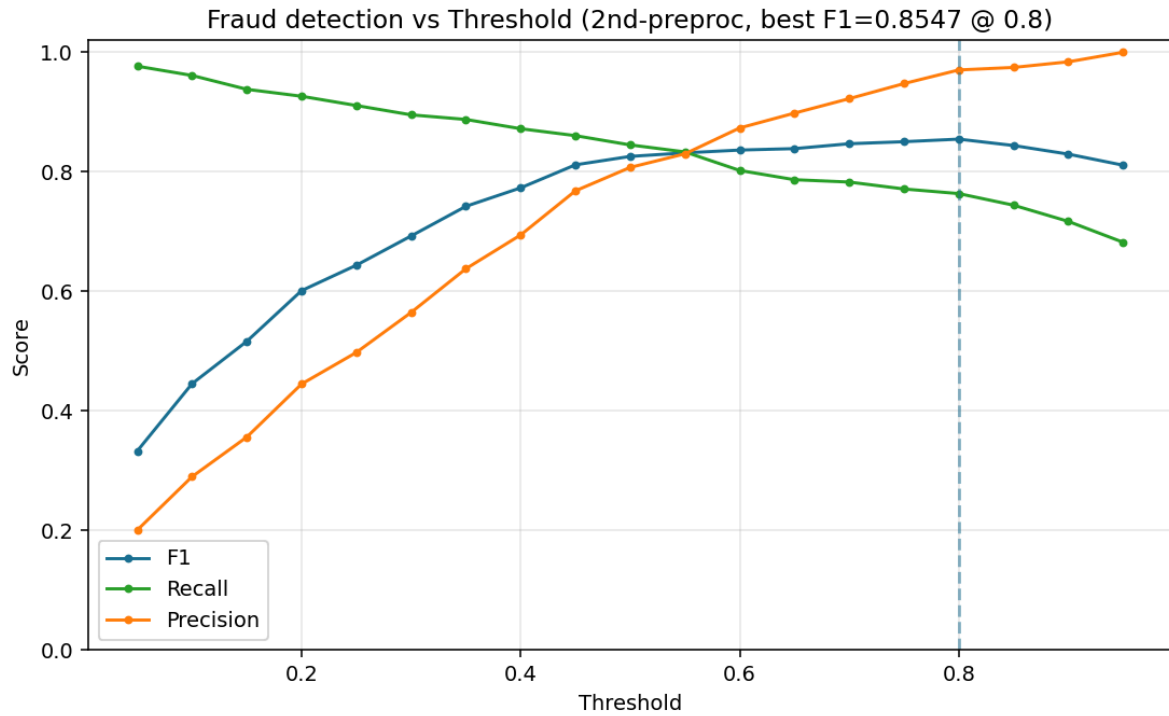
feature	mean_abs_shap
Transaction_is_withdrawal	0.3156
Channel	0.2155
Transaction_Amount_abs	0.1626
Amount_vs_monthly_max_ratio	0.1583
Account_one_month_std_dev	0.1582
Flag_deposit_more_than_tenMillion	0.1569
Customer_Birthyear	0.1142
Account_release_suspention	0.0975
Access_Medium	0.0902
Distance	0.0854
Account_account_type	0.0845
Time_difference_seconds	0.0681
Transaction_history_with_the_account	0.0673
Amount_vs_remaining_balance	0.0638
Customer_credit_rating	0.0613

혼동행렬



	pred_a	pred_b	pred_c	pred_d	pred_e	pred_f	pred_g
actual_a	21	0	0	1	0	0	0
actual_b	1	16	0	0	0	0	0
actual_c	0	0	7	2	2	0	1
actual_d	0	1	1	5	1	0	1
actual_e	1	0	0	0	11	0	1
actual_f	0	0	0	0	0	12	1
actual_g	0	0	1	3	2	4	9
actual_h	0	0	0	1	0	0	1
actual_i	0	0	0	1	0	0	0
actual_j	0	0	1	2	0	0	1
actual_k	0	1	0	1	2	0	0
actual_l	0	0	1	0	3	1	0

임계값별_F1



임계값	Recall	Precision	F1	미탐(FN)	오탐(FP)
0.05	0.9767	0.2008	0.3331	6	1003
0.1	0.9612	0.2901	0.4456	10	607
0.15	0.938	0.3564	0.5165	16	437
0.2	0.9264	0.4451	0.6013	19	298
0.25	0.9109	0.4979	0.6438	23	237
0.3	0.8953	0.5648	0.6927	27	178
0.35	0.8876	0.6379	0.7423	29	130
0.4	0.8721	0.6944	0.7732	33	99
0.45	0.8605	0.7682	0.8117	36	67
0.5	0.845	0.8074	0.8258	40	52
0.55	0.8333	0.8301	0.8317	43	44
0.6	0.8023	0.8734	0.8364	51	30
0.65	0.7868	0.8982	0.8388	55	23
0.7	0.7829	0.9224	0.847	56	17
0.75	0.7713	0.9476	0.8504	59	11
0.8	0.7636	0.9704	0.8547	61	6
0.85	0.7442	0.9746	0.844	66	5
0.9	0.7171	0.984	0.8296	73	3
0.95	0.6822	1.0	0.8111	82	0

최종 결론 : 최종 성능 = Macro-F1 0.578 (is_fraud 제거 기준) 정상 vs 사기 임계값 0.8 에서 F1 0.855